

and Lie algebras [127] [135] [289] [96, Sect. 2.3]. There is a conjecture that except for a few “obvious” infinite families, most of which come from these theories, there are only finitely many “sporadic” examples: but currently no one seems to have the faintest idea how to prove this. We refer to work by Grünbaum [254] for the case $r = 3$, and to Grünbaum & Shephard [258] for the case $r = 4$. The enumeration of all “known” arrangements of rank 3 attempted in [254] had only one addition and one correction up to now (Grünbaum [255], and Barthel, Hirzebruch & Höfer [56]) and might be essentially complete, while the enumeration of [258] for $r \geq 4$ is probably far from complete; see Alexanderson & Wetzel [8, 9, 10]. Up-to-date references can be found in Wetzel [556].

The Bohné-Dress theorem was announced by Andreas Dress at the 1989 Symposium on Combinatorics and Geometry in Stockholm. It is a strikingly simple geometric observation that had previously eluded people. A complete proof, however, is surprisingly difficult, and it took some time until the complete written version by Bohné [111] was available. A simpler, more geometric proof is given by Richter-Gebert & Ziegler [461]. Our sketch in Section 7.5 follows that paper.

The Bohné-Dress theorem relates the set of all zonotopal tilings on a given zonotope with an extension space problem (“Is the space of all extensions of an oriented matroid homotopy equivalent to a sphere?”, see Sturmfels & Ziegler [536]). Thus zonotopal tilings allow one to study a special case of two very basic, general, and apparently very difficult problems, the “generalized Baues problem” of Billera, Kapranov & Sturmfels [73], and the problem of “Combinatorial Grassmannians” by MacPherson [373], see also in Mnëv & Ziegler [410]. We will discuss the setting of the Generalized Baues Problem in Lecture 9. For the problems themselves and their ramifications we refer to the original sources.

Problems and Exercises

- 7.0 In Definition 7.1, show that if the cones are convex, then they are automatically polyhedral, so the condition “polyhedral” could be dropped from the definition.
- 7.1 Let $P \subseteq \mathbb{R}^d$ be a polytope with $\mathbf{0}$ in its interior. Show that the face fan of P is the normal fan of the polar P^Δ , and the normal fan of P is the face fan of P^Δ .
- 7.2 Enumerate all the 3-zonotopes generated by 5 vectors in $\mathbb{R}^3$, draw them, and count the vertices, facets, and simple vertices.
What about 6 vectors in $\mathbb{R}^3$? What about 5 or 6 vectors in $\mathbb{R}^4$?
- 7.3 Give examples of simple zonotopes and of simple zonotopal tilings.

7.4 Show that the following are equivalent for a polytope Z :

- every 2-face of Z has an even number of edges, and opposite edges are parallel.
- for every edge, Z has some multiple or part of it as a Minkowski summand.
- the normal fan of Z is a hyperplane arrangement.

(Such polytopes, *generalized zonotopes*, were introduced by the Russian crystallographer Fedorov; Coxeter apparently misunderstood the definition and assumed that Fedorov was considering zonotopes — see Taylor [538]. Bolker [119] studies them under the name of *planets*, Baladze [36] calls them *belt polytopes*.)

Give examples of generalized zonotopes that are not zonotopes. The above shows, however, that every generalized zonotope is combinatorially equivalent (in fact, normally equivalent) to a zonotope.

7.5 If every projection of a polytope to $\mathbb{R}^3$ is a zonotope, then the polytope is a zonotope itself.

Show that the projections to $\mathbb{R}^2$ are not good enough for this. (Witsenhausen [568])

7.6 Interpret the deletion and the contraction of a vector in the configuration V in terms of zonotopes. That is, describe how the zonotopes $Z(V \setminus \mathbf{v})$ and $Z(V/\mathbf{v})$ can be constructed geometrically.

7.7 Let $Z(V)$ be the zonotope generated by a configuration $V \in \mathbb{R}^{d \times n}$ which spans $\mathbb{R}^d$. Let $G \in (\mathbb{R}^*)^{(n-d) \times n}$ be the dual vector configuration.

- (i) How can the combinatorics of the zonotope $Z(V)$ be read off from the configuration G ?
- (ii) Use this to describe the zonotopes with $n \leq d + 2$ zones.
- (iii) Describe the relation between the zonotope $Z(V)$ and its *associated zonotope* $Z(G) \subseteq (\mathbb{R}^{n-d})^*$.

(This was developed by McMullen [392].)

7.8 Assume that $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ are given. Give an explicit formula for some small enough $\varepsilon > 0$ such that $\text{sign}(\mathbf{x} + \varepsilon \mathbf{y}) = \text{sign}(\mathbf{x}) \circ \text{sign}(\mathbf{y})$.

7.9 Let $\mathcal{V}^* \subseteq \{+, -, 0\}^n$ be a system of sign vectors.

- (i) Assuming that (V0): $\mathbf{0} \in \mathcal{V}^*$ holds, show that the axioms (V1) and (V2) together are equivalent to the axiom
(V2'): $\mathbf{u}, \mathbf{v} \in \mathcal{V}^* \implies \mathbf{u} \circ (-\mathbf{v}) \in \mathcal{V}^*$.

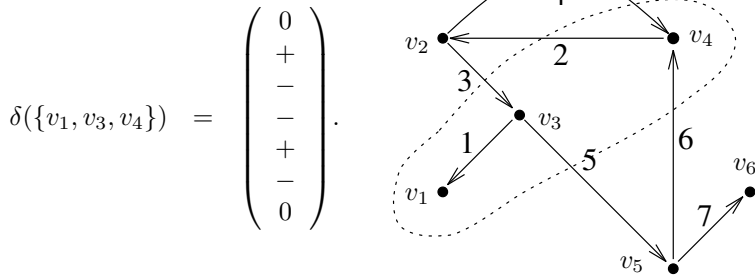
- (ii) Consider any affine arrangement of n hyperplanes in which a positive side has been chosen for each of the hyperplanes. Show how a sign vector is associated with every face of the arrangement, and the resulting collection of sign vectors satisfies (V2') and (V3), but not in general (V0) and (V1).
- (iii) Show that not every sign vector system satisfying (V2') and (V3) corresponds to an affine arrangement, even if we admit topologically deformed arrangements ("pseudoarrangements").

(The characterization of the sign vector systems of affine oriented matroids was a difficult combinatorial problem, recently solved by Karlander [316].)

- 7.10 Let $D = (V, A)$ be a directed graph with n arcs, $A = \{a_1, \dots, a_n\}$. With every subset $U \subseteq V$ of its vertex set, associate a sign vector $\delta(U) \in \{+, -, 0\}^n$, the "directed cut" of U , where $\delta(U)_i = +$ if a_i is an arc that leaves the set U ; $\delta(U)_i = -$ if a_i enters the set U ; and $\delta(U)_i = 0$ otherwise:

$$\delta(U)_i = \begin{cases} + & \text{if } \text{tail}(a_i) \in U \text{ and } \text{head}(a_i) \notin U, \\ - & \text{if } \text{tail}(a_i) \notin U \text{ and } \text{head}(a_i) \in U, \\ 0 & \text{if } \text{tail}(a_i) \text{ and } \text{head}(a_i) \text{ are both in } U, \\ & \text{or both not in } U. \end{cases}$$

For example, in the graph we have drawn



Show that the family $\mathcal{V}^* := \{\delta(U) : U \subseteq V\}$ is an oriented matroid. What is its rank?

What is the relation to the oriented matroid associated with such a digraph according to Exercise 6.3?

- 7.11 Show directly from the axioms in Definition 7.21 that every oriented matroid of rank $r \leq 2$ (that is, an oriented matroid $\mathcal{V}^* \subseteq \{+, -, 0\}^n$ that does not contain a chain $\mathbf{0} < X < X' < X''$) is realizable.
- 7.12 Show the following theorem.

Let $\mathcal{V}^*, \mathcal{C}^* \subseteq \{+, -, 0\}^n$ be sign vector systems such that $\mathcal{C}^*$ is the collection of sign vectors of minimal nonempty support in $\mathcal{V}^*$,

$$\mathcal{C}^* = \text{MIN}(\mathcal{V}^*),$$

while $\mathcal{V}^*$ is the collection of all conformal products of sign vectors in $\mathcal{C}^*$,

$$\mathcal{V}^* = \{\mathbf{0} \circ v_1 \circ \dots \circ v_k : k \geq 0, v_i \in \mathcal{C}^* \text{ for } i = 1, 2, \dots, k\}.$$

Then $\mathcal{V}^*$ is the covector set of an oriented matroid (that is, it satisfies the axioms of Definition 7.21) if and only if $\mathcal{C}^*$ is the set of *cocircuits of an oriented matroid*, that is, if $\mathcal{C}^*$ satisfies the following *cocircuit axioms*:

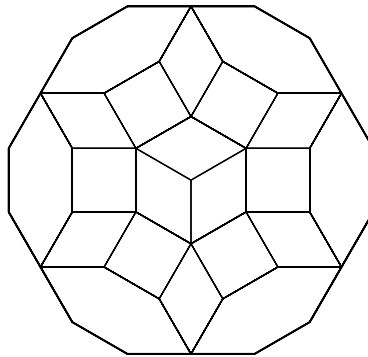
- (C0) $\mathbf{0} \notin \mathcal{C}^*$ (“The zero vector is not a cocircuit”)
- (C1) $u \in \mathcal{C}^* \implies -u \in \mathcal{C}^*$
 (“The negative of a cocircuit is always a cocircuit”)
- (C2) $u, v \in \mathcal{C}^*, \text{supp}(u) \subseteq \text{supp}(v) \implies u = \pm v$
 (“Cocircuits have noncomparable supports”)
- (C3) $u, v \in \mathcal{C}^*, u \neq -v, j \in S(u, v) \implies \exists w \in \mathcal{C}^*, w' \in \{+, -, 0\}^n$:
 $w \leq w'$, and w' eliminates j between u and v (“The set of cocircuits admits elimination”)

7.13 Show that for any vector subspace $U \subseteq \mathbb{R}^n$, the set of minimal nonempty supports in $\mathcal{U} = \text{SIGN}(U)$, given by $\text{MIN}(\text{SIGN}(U)) = \{u \in \text{sign}(U) \setminus \{\mathbf{0}\} : v \in \text{sign}(U), \text{supp}(v) \subset \text{supp}(u) \text{ implies } v = \mathbf{0}\}$, is the set of cocircuits of an oriented matroid, that is, it satisfies the axioms of Exercise 7.12.

7.14 How can you test whether a given zonotopal tiling is the picture of an actual zonotope? Show that, essentially, one has to decide whether a certain polyhedron has nonempty interior, which can be solved as a linear programming problem, as in Exercise 5.2(i).
 So, is it true that the first figure in Section 7.5 represents the picture of a 3-dimensional zonotope?

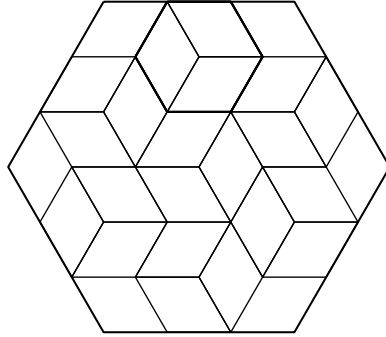
7.15 Show that every nontrivial zonotopal tiling in $\mathbb{R}^2$ (a tiling of a centrally symmetric polygon by centrally symmetric polygons) has a simple vertex on the boundary, and also a simple vertex in the interior.

7.16 Consider the following zonotopal tiling.



Show that it corresponds to an arrangement of 9 pseudolines which is not stretchable, because it violates Desargues' theorem.

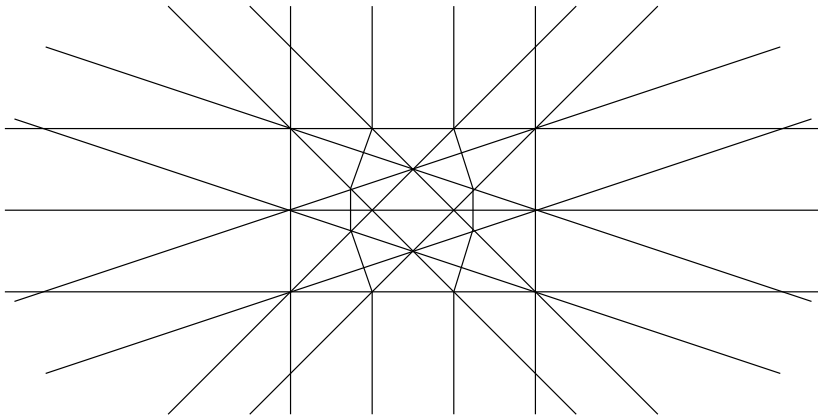
Show that the pseudoline arrangement of the following tiling is also not stretchable:



(The second configuration is closely related to Ringel's simple configuration of 9 pseudolines; one only has to delete the line at infinity and perturb the arrangement there.)

These drawings were produced by Jürgen Richter-Gebert, using his postscript program described and listed in [457], which produces exceptionally nice pictures of zonotopal tilings.)

- 7.17 Consider the following simplicial arrangement of 16 pseudolines from Grünbaum [255, p. 44].



- (i) Show that it is not realizable.
- (ii) Use it to construct a simple zonotopal tiling of a 12-gon whose oriented matroid is not realizable.

7.18 For every d -zonotope, the numbers f_k of k -faces satisfy the relations

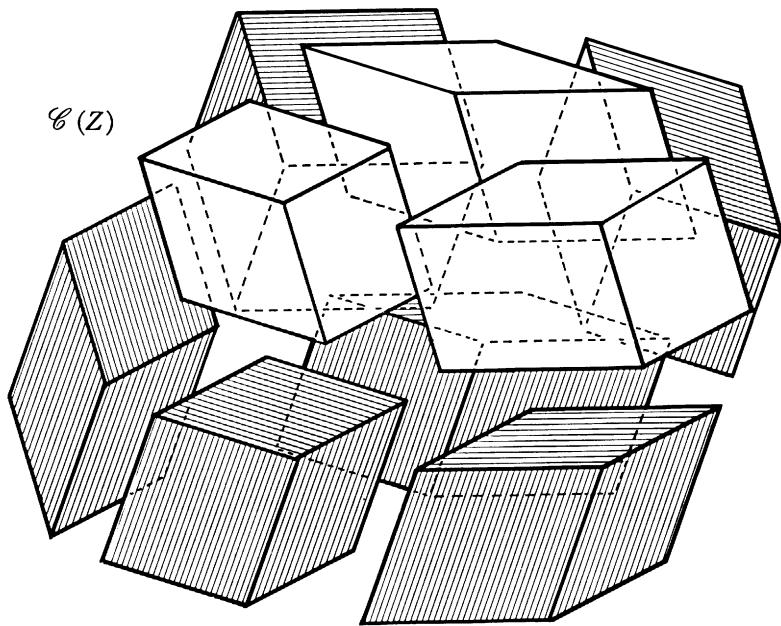
$$f_{k-1} \geq \frac{k}{d-k+1} f_k \text{ for } 1 \leq k \leq d, \text{ and thus } f_k \leq \binom{d}{k} f_0.$$

(This is given in terms of hyperplane arrangements and oriented matroids in Fukuda, Tamura & Tokuyama [214, 213].)

7.19 For any vector configuration $V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subseteq \mathbb{R}^d$, prove the volume formula for its zonotope:

$$\text{vol}(Z(V)) = 2^d \cdot \sum_{1 \leq i_1 < \dots < i_d \leq n} |\det(\mathbf{v}_{i_1}, \dots, \mathbf{v}_{i_d})|.$$

For this, decompose the zonotopes into parallelepipeds, whose volumes are given by determinants.



(McMullen, see Shephard [495, Sect. 5], from where we have also taken the illustration.)

7.20* You can use the formula in the preceding exercise to compute the volume of a zonotope, but that is not very effective: the formula has $\binom{n}{d}$ terms, which may all be nonzero.

Is there a fast (polynomial) way to compute the volume of a zonotope? (Answer: most probably not — this is #P-hard according to Dyer, Gritzmann & Hufnagel [187].)

8

Shellability and the Upper Bound Theorem

Perhaps the most famous result about convex polytopes is the *Euler-Poincaré formula*:

$$-f_{-1} + f_0 - f_1 + f_2 + \cdots + (-1)^{d-1}f_{d-1} + (-1)^df_d = 0,$$

where f_i denotes the number of i -dimensional faces of a d -polytope P . Here $f_{-1} = 1$ and $f_d = 1$ correspond to the trivial faces (the empty face and the polytope itself), and f_0 , f_1 , f_{d-2} , and f_{d-1} are the numbers of vertices, edges, ridges, and facets, respectively. So for 2-polytopes we obtain that $f_0 - f_1 = 0$, the number of vertices equals the number of edges (not much of a surprise). For 3-polytopes we get “Euler’s formula”

$$v - e + f = 2$$

for a 3-polytope with $v = f_0$ vertices, $e = f_1$ edges, and $f = f_2$ facets.

For $d \leq 3$ the Euler-Poincaré formula is easy to prove, but for higher dimensions care is needed. As Grünbaum [252] observed, all the classical inductive proofs (starting with Schläfli’s [473] proof from 1852; see also Sommerville [506, p. 147], Schoute [480, p. 61], and the references in [252, p. 141]) assume that the boundary of every polytope can be built up inductively in a nice way, that is, it can be “shelled.” That this is in fact possible was only proved by Bruggesser & Mani in 1970. A striking application of shellability was McMullen’s proof of the “upper bound theorem” in the same year, 1970.

In this lecture we have several big goals. We start with shellability for polyhedral complexes, a concept that is both useful and nontrivial. We will show that